Arista Access Node SOP: ALARM_9476 - Optical Transport Failure (JAY-ACC-55)

Node: JAY-ACC-55 | **Vendor:** Arista | **Severity:** Minor | **Alarm Code:** ALARM_9476

Description

The Arista Access Node at Jayanagar Metro Site (Node ID: JAY-ACC-55) is experiencing an Optical Transport Failure (Alarm Code: ALARM_9476), categorized as Minor Severity. This failure domain pertains to the degradation of SFP/XFP transceivers, fiber cuts, or Rx/Tx power loss, which are inherent to the Optical Transport Network (OTN) architecture. The OTN is a high-speed networking protocol that enables the transport of data over optical fibers. On Arista equipment, specifically in the Access role, the OTN protocol is utilized to interconnect multiple devices and provide high-speed connectivity. The failure of SFP/XFP transceivers or fiber cuts within the OTN network can lead to data transmission disruptions, resulting in the ALARM_9476 condition. This SOP outlines the procedures to diagnose and resolve the issue.

Pre-requisite Checks

- Verify that the Jayanagar Metro Site is accessible for remote hands dispatch.
- Check the redundant link status to ensure it is operational and not contributing to the failure.
- Gather relevant site documentation, including the equipment configuration and maintenance records.

Potential Root Causes

Cause	Probability
SFP/XFP Transceiver Degradation	High
Fiber Cut or Damage	Medium
Rx/Tx Power Loss	Low

Diagnostic Commands & Expected Outputs

```
Command: show interface detail | include 'rx bytes' 'tx bytes', Expected Output: rx bytes: 1234567890/1234567890, tx bytes: 1234567890/1234567890, Description: This command isolates the receive and transmit bytes counters for the affected interface, helping to determine if data transmission is occurring and if the issue is related to data loss or corruption.

Command: show optics details | include 'temperature' 'voltage', Expected Output: temperature: 25.0C, voltage: 3.3V, Description: This command provides information about the optical transceiver's temperature and voltage, which can help identify potential issues related to environmental conditions or hardware degradation.
```

```
Command: show optics statistics | include 'error-counts', Expected Output: error-counts: 0x00000000, Description: This command displays error counts for the optical transceiver, which can indicate issues with data transmission or reception.
```

Resolution Steps

Step 1: Step: Replace SFP/XFP Transceiver, Command: replace optics 1/1/1, Description: This command initiates the replacement of the faulty SFP/XFP transceiver with a new one. Ensure the new transceiver is compatible with the existing equipment and configuration.

Step 2: Step: Verify Fiber Integrity, Command: show optics details | include 'fiber-status', Description: This command checks the status of the fiber connection, including any potential damage or cuts. If the fiber is damaged, replace it with a new one.

Step 3: Step: Adjust Rx/Tx Power Levels, Command: configure optics 1/1/1 rx-power 3.3V tx-power 3.3V, Description: This command adjusts the receive and transmit power levels for the optical transceiver to optimal values. Refer to the equipment documentation for recommended power levels.

Post-Check Verification

```
Command: show interface detail | include 'rx bytes' 'tx bytes', Expected Output: rx bytes: 1234567890/1234567890, tx bytes: 1234567890/1234567890, Description: Verify that data transmission is occurring without issues by checking the receive and transmit bytes counters.

Command: show optics details | include 'temperature' 'voltage', Expected Output: temperature: 25.0C, voltage: 3.3V, Description: Confirm that the optical transceiver's temperature and voltage are within acceptable ranges.

Command: show optics statistics | include 'error-counts', Expected Output: error-counts: 0x00000000, Description: Check that error counts for the optical transceiver are zero, indicating successful data transmission.
```

Roll-back Procedures

- Step: Restore Original SFP/XFP Transceiver, Command: replace optics 1/1/1 original, Description: If the replacement SFP/XFP transceiver did not resolve the issue, restore the original transceiver to its previous state.

- Step: Revert Fiber Configuration, Command: configure optics 1/1/1 fiber-status original, Description: If the replacement fiber did not resolve the issue, revert the fiber configuration to its previous state.

- Step: Revert Rx/Tx Power Levels, Command: configure optics 1/1/1 rx-power original tx-power original, Description: If the adjusted Rx/Tx power levels did not resolve the issue, revert the power levels to their original values.